📚 Complete Data Structures Interview Questions Guide

1. 🌳 **TREES** (Binary Trees, BST, AVL, etc.)

Easy Questions

```cpp
1. Implement tree traversals (inorder, preorder, postorder, level order)
2. Find height/depth of a tree
3. Count total nodes, leaf nodes, non-leaf nodes
4. Search for a value in BST
5. Find minimum and maximum in BST
6. Check if two trees are identical
7. Print nodes at given level
8. Implement level order traversal using queue
9. Find size of a tree
10. Check if a node exists in tree
```

Medium Questions

```cpp
11. Check if a tree is BST
12. Find Lowest Common Ancestor (LCA)
13. Print left/right/top/bottom view
14. Find diameter of tree
15. Check if tree is balanced
16. Find mirror of tree
17. Check if two trees are mirrors
18. Print all root-to-leaf paths
```

19. Find path sum equals target

20. Convert tree to its mirror

21. Find inorder successor/predecessor

22. Delete a node from BST

23. Construct tree from inorder+preorder

24. Construct tree from inorder+postorder

25. Serialize and deserialize tree (like your save/load!)

```

### Hard Questions

```cpp

26. Find maximum path sum (any node to any node)

27. Flatten tree to linked list

28. Convert BST to balanced BST

29. Find kth smallest/largest in BST

30. Merge two BSTs

31. Find median in BST

32. Count pairs from two BSTs with given sum

33. Check if array represents preorder of BST

34. Recover BST (two nodes swapped)

35. Largest BST in binary tree

36. Tree from preorder and postorder

37. Morris traversal (O(1) space)

38. Binary Tree to DLL in-place

39. Find distance between two nodes

40. Print nodes at distance k from target

```
```

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2. 📊 **ARRAYS**

Easy Questions

```cpp
1. Reverse an array
2. Find min and max in array
3. Find sum of all elements
4. Find missing number (1 to n)
5. Check if array is sorted
6. Remove duplicates from sorted array
7. Rotate array by k positions
8. Move all zeros to end
9. Find second largest element
10. Count frequency of elements
```

Medium Questions

```cpp
11. Find two numbers with given sum (2 Sum)
12. Find majority element (> n/2 times)
13. Find leaders in array (greater than right)
14. Rearrange positive and negative
15. Find subarray with given sum
16. Maximum subarray sum (Kadane's)
17. Merge two sorted arrays
18. Find intersection of two arrays
```

19. Find union of two arrays

20. Find duplicates in array

21. Trapping rain water

22. Stock buy and sell (max profit)

23. Find triplet with given sum (3 Sum)

24. Find if array can be partitioned

25. Container with most water

```

### Hard Questions

```cpp

26. Find longest consecutive subsequence

27. 4 Sum problem

28. Maximum product subarray

29. Find median of two sorted arrays

30. Minimum swaps to sort array

31. Find first missing positive

32. Find duplicate number (without extra space)

33. Merge overlapping intervals

34. Insert interval in sorted intervals

35. Find minimum in rotated sorted array

36. Search in rotated sorted array

37. Find peak element

38. Count inversions in array

39. Maximum circular subarray sum

40. Jump game (minimum jumps)

```
```

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3. 🔗 **LINKED LISTS**

Easy Questions

```cpp
1. Create and traverse linked list
2. Insert at beginning/end/position
3. Delete from beginning/end/position
4. Find length of linked list
5. Search for element
6. Reverse a linked list
7. Find middle element
8. Find nth node from end
9. Count occurrences of value
10. Check if linked list is palindrome
```

Medium Questions

```cpp
11. Detect loop in linked list (Floyd's cycle)
12. Find start of loop
13. Remove loop from linked list
14. Merge two sorted linked lists
15. Intersection point of two lists
16. Add two numbers represented by lists
17. Remove duplicates from sorted list
18. Remove duplicates from unsorted list
```

19. Partition list around value

20. Rotate list by k positions

21. Split circular linked list

22. Check if linked list is circular

23. Sort linked list (merge sort)

24. Reverse in groups of k

25. Flatten multi-level linked list

```

### Hard Questions

```cpp

26. Clone linked list with random pointer

27. LRU Cache implementation

28. Merge k sorted linked lists

29. Rearrange linked list (alternating)

30. Palindrome in O(1) space

31. Subtract two numbers represented by lists

32. Quick sort on linked list

33. Multiply two numbers represented by lists

34. Next greater node in linked list

35. Linked list in zig-zag pattern

36. Swap nodes in pairs

37. Even odd linked list

38. Delete nodes with greater value on right

39. Sort linked list of 0s, 1s, 2s

40. Merge sort on doubly linked list

```
```

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4. 📦 **STACKS**

Easy Questions

```cpp
1. Implement stack using array
2. Implement stack using linked list
3. Push, pop, peek operations
4. Check if parentheses are balanced
5. Reverse a string using stack
6. Implement two stacks in one array
7. Sort a stack (using recursion)
8. Find middle element of stack
9. Check if expression has redundant brackets
10. Next greater element (simpler version)
```

Medium Questions

```cpp
11. Next greater element (all versions)
12. Next smaller element
13. Previous greater/smaller element
14. Largest rectangle in histogram
15. Stock span problem
16. Evaluate postfix expression
17. Convert infix to postfix
18. Convert infix to prefix
```

19. Design min stack (getMin in O(1))

20. Validate stack sequences

21. Decode string (like "3[a2[c]]")

22. Remove adjacent duplicates

23. Check if stack is palindrome

24. Reverse stack without extra space

25. Maximum area in binary matrix

```

### Hard Questions

```cpp

26. Maximum of minimums for every window

27. Celebrity problem

28. The celebrity problem using stack

29. Length of longest valid parentheses

30. Trapping rain water (stack approach)

31. Expression contains redundant bracket

32. Simplify path (Unix style)

33. Basic calculator I, II, III

34. Parsing boolean expression

35. Remove k digits to form smallest number

36. Asteroid collision

37. Car fleet problem

38. Maximum Nested Depth of Parentheses

39. Minimum add to make parentheses valid

40. Score of parentheses

```
```

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5. 🔄 **QUEUES**

Easy Questions

```cpp
1. Implement queue using array
2. Implement queue using linked list
3. Enqueue, dequeue operations
4. Implement circular queue
5. Generate binary numbers from 1 to n
6. Reverse a queue
7. Implement stack using queues
8. Implement queue using stacks
9. First non-repeating character in stream
10. Check if queue is palindrome
```

Medium Questions

```cpp
11. Sliding window maximum
12. Implement Deque (double-ended queue)
13. LRU Cache using queue
14. Rotten oranges problem
15. Time needed to buy tickets
16. Design circular deque
17. Reveal cards in increasing order
18. Dota2 senate problem
```

19. Queue based approach for BFS

20. Level order traversal using queue

21. Zigzag level order traversal

22. Connect nodes at same level

23. Minimum time to rot all oranges

24. Jump game III (BFS approach)

25. Snake and ladder problem

```

### Hard Questions

```cpp

26. Task scheduler

27. Rearrange string k distance apart

28. Design hit counter

29. Moving average from data stream

30. Find median from data stream (two heaps)

31. Maximum sum of k non-overlapping subarrays

32. Shortest path in binary matrix (BFS)

33. Word ladder I and II

34. Open the lock (BFS problem)

35. Sliding window median

36. Minimum cost to connect ropes

37. Reorganize string

38. Maximum width of binary tree

39. Minimum moves to move boxes

40. The maze (BFS/DFS problem)

```
```

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6. 🔤 **STRINGS**

Easy Questions

```cpp
1. Reverse a string
2. Check palindrome
3. Count vowels and consonants
4. Count frequency of characters
5. Remove duplicates from string
6. Check anagrams
7. Convert to uppercase/lowercase
8. Find first non-repeating character
9. Check if string contains only digits
10. Longest common prefix
```

Medium Questions

```cpp
11. Longest substring without repeating chars
12. Longest palindromic substring
13. String to Integer (atoi)
14. Implement strstr (pattern matching)
15. Group anagrams
16. Valid parentheses
17. Generate all permutations of string
18. Count and say sequence
```

19. Zigzag conversion

20. Multiply strings

21. Add two binary strings

22. Valid palindrome (with non-alphanumeric)

23. Reverse words in a string

24. Compare version numbers

25. Minimum window substring

```

### Hard Questions

```cpp

26. Regular expression matching

27. Wildcard matching

28. Edit distance (Levenshtein)

29. Distinct subsequences

30. Word break I and II

31. Palindrome partitioning

32. Scramble string

33. Minimum insertions to make palindrome

34. Decode ways

35. Encode and decode strings

36. Text justification

37. Valid number (lexicographic)

38. Interleaving strings

39. Longest valid parentheses

40. Shortest palindrome

41. KMP Algorithm (pattern search)

42. Rabin-Karp (rolling hash)

43. Z algorithm

44. Manacher's algorithm

45. Suffix array/automaton

```

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7. 🌐 **GRAPHS**

Easy Questions

```cpp

1. Implement graph using adjacency matrix/list

2. BFS traversal

3. DFS traversal

4. Detect cycle in undirected graph (DFS)

5. Count connected components

6. Check if path exists between nodes

7. Find degree of a node

8. Check if graph is bipartite

9. Find number of nodes at given level

10. Transpose of graph

```

Medium Questions

```cpp

11. Detect cycle in directed graph

12. Topological sorting

13. Shortest path in unweighted graph (BFS)

14. Dijkstra's algorithm (shortest path)

15. Bellman-Ford algorithm

16. Floyd Warshall (all pairs shortest)

17. Prim's MST (Minimum Spanning Tree)

18. Kruskal's MST

19. Number of islands (2D grid)

20. Rotting oranges (BFS)

21. Clone graph

22. Course schedule (cycle detection)

23. Alien dictionary

24. Evaluate division equations

25. Find if path exists in grid

26. Snakes and ladders

27. Word ladder

28. Pacific Atlantic water flow

29. Surrounded regions

30. Walls and gates

```

Hard Questions

```cpp

31. Strongly connected components (Kosaraju/Tarjan)

32. Articulation points (cut vertices)

33. Bridges in graph

34. Travelling salesman problem

35. Hamiltonian path/cycle

36. Graph coloring

37. Maximum flow (Ford-Fulkerson)

38. Min cut (Max flow min cut)

39. Bipartite matching

40. Job assignment problem

41. Critical connections in network

42. Reconstruct itinerary

43. Minimum height trees

44. Longest increasing path in matrix

45. Word search II (Trie + DFS)

46. Remove invalid parentheses

47. Shortest path in grid with obstacles

48. Minimum obstacles to remove

49. Swim in rising water

50. Cheapest flights within k stops

51. Network delay time

52. Path with maximum probability

53. Find eventual safe states

54. Possible bipartition

55. Minimize malware spread

```

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8. 💎 **HASHING**

Easy Questions

```cpp

1. Implement hash table with chaining

2. Implement hash table with open addressing

3. Count frequency of elements

4. Find first repeating element

5. Find symmetric pairs in array

6. Check if array contains duplicates

7. Two Sum using hash map

8. Find intersection of two arrays

9. Find union of two arrays

10. Check if two arrays are equal

```

Medium Questions

```cpp

11. Longest consecutive sequence

12. Subarray sum equals K

13. Group anagrams

14. Find all duplicates in array

15. Four sum problem

16. Largest subarray with zero sum

17. Count subarrays with given XOR

18. Find itinerary from tickets

19. Isomorphic strings

20. Word pattern matching

21. Fraction to recurring decimal

22. Random pool (insert, delete, random)

23. Design HashMap with O(1) operations

24. Minimum window substring

25. Longest substring with at most k distinct chars

26. Palindrome pairs

27. Count distinct elements in window

28. Find pairs with given sum

29. Check if array can be divided into pairs

30. Sort characters by frequency

```

Hard Questions

```cpp

31. Design LRU cache

32. Design LFU cache

33. Maximum frequency stack

34. Design Twitter feed

35. Find all subarrays with given sum

36. Count of range sum

37. Rabbit in forest (min possible rabbits)

38. Brick wall (vertical line cutting)

39. Distribute candies (fair distribution)

40. K-diff pairs in array

41. Subdomain visit count

42. Find duplicate file in system

43. Most frequent subtree sum

44. Beautiful array (hash + recursion)

45. Powerful integers

46. Count number of subarrays with max in range

47. Find k-length substrings with no repeats

48. Pairs of songs with total durations divisible by 60

49. Alert using same key card

50. Display table of food orders

```

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9. ⚡ **HEAPS / PRIORITY QUEUES**

Easy Questions

```cpp
1. Implement min heap
2. Implement max heap
3. Heapify an array
4. Find k largest elements
5. Find k smallest elements
6. Sort a nearly sorted array
7. Kth largest element in array
8. Kth smallest element in array
9. Merge k sorted arrays
10. Check if array represents heap
```

Medium Questions

```cpp
11. Find median from data stream
12. Top k frequent elements
13. Task scheduler (min CPU intervals)
14. Reorganize string (no two adjacent same)
15. Connect n ropes with minimum cost
16. Sliding window maximum
```
```

17. K closest points to origin

18. Maximum distinct elements after removal

19. Minimum cost to hire k workers

20. Ugly number II (heap approach)

21. Find k pairs with smallest sums

22. Rearrange string k distance apart

23. Sort characters by frequency

24. Minimum sum of two numbers from array

25. Maximum product of k numbers

```

### Hard Questions

```cpp

26. Maximum sum combinations from two arrays

27. Find k smallest sums with one from each array

28. Minimum number of refueling stops

29. Trapping rain water II (3D)

30. The skyline problem

31. Employee free time

32. Meeting rooms III

33. Minimum cost to connect all points (Prim's)

34. Process tasks using servers

35. Maximum performance of a team

36. Find kth smallest in sorted matrix

37. Swim in rising water (Dijkstra + heap)

38. Path with minimum effort

39. Minimum obstacle removal to reach corner

40. Minimum time to visit all points

41. Course schedule III (deadlines)

42. Maximum profit in job scheduling

43. Car pooling (capacity check)

44. Split array into consecutive subsequences

45. Remove stones to minimize total

46. Reduce array size to half

47. Least number of unique integers after removal

48. Maximum number of events that can be attended

49. Single-threaded CPU

50. Find kth largest XOR coordinate value

```

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10. 🔍 **SEARCHING & SORTING**

Easy Questions

```cpp

1. Linear search

2. Binary search (iterative & recursive)

3. First and last occurrence in sorted array

4. Count occurrences in sorted array

5. Bubble sort

6. Selection sort

7. Insertion sort

8. Check if array is sorted and rotated

9. Find peak element

10. Search in infinite sorted array

```

Medium Questions

```cpp
11. Merge sort
12. Quick sort
13. Count sort
14. Search in rotated sorted array
15. Find minimum in rotated sorted array
16. Search in nearly sorted array
17. Find floor/ceil of element in sorted array
18. Find position of element in infinite array
19. Find missing element in AP
20. Find peak in 2D array
21. Find majority element (Moore's voting)
22. Sort array of 0s, 1s, 2s (Dutch flag)
23. Find pair with given difference
24. Find triplet with sum zero
25. Find common elements in three sorted arrays
26. Sort array by parity
27. Find k closest elements
28. Maximum gap (bucket sort)
29. Sort array based on frequency
30. Wave sort
```

Hard Questions

```cpp

31. Quick select (kth smallest/largest)

32. Median of two sorted arrays

33. Aggressive cows (min distance max)

34. Allocate minimum pages

35. Painter's partition problem

36. Search in row and column wise sorted matrix

37. Find median in row-wise sorted matrix

38. Find kth smallest in sorted matrix

39. Count of smaller numbers after self

40. Count inversions (merge sort)

41. Maximum sum of k non-overlapping subarrays

42. Minimum swaps to sort array

43. First missing positive

44. Find duplicate number (binary search)

45. Split array largest sum

46. Find in mountain array

47. Capacity to ship packages within D days

48. Koko eating bananas

49. Minimum time to complete trips

50. Maximum candies allocated to k children

51. Magnetic force between two balls

52. Minimum limit of balls in a bag

53. Find the smallest divisor given a threshold

54. Nth magical number

55. Swim in rising water

56. Path with minimum effort

57. Find kth smallest pair distance

58. Find kth smallest number with multiplication table

59. Find kth smallest sum of matrix with sorted rows

60. Maximize sweetness of candy

```

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11. 🔄 **RECURSION & BACKTRACKING**

Easy Questions

```cpp

1. Factorial of a number

2. Fibonacci series

3. Sum of digits

4. Power of a number (x^n)

5. Check palindrome using recursion

6. Reverse a string using recursion

7. Print 1 to n without loops

8. Tower of Hanoi

9. Generate all subsets (powerset)

10. Permutations of string (simple)

```

Medium Questions

```cpp

11. N-Queens problem

12. Sudoku solver

13. Rat in a maze

14. Knight's tour

15. Generate all permutations

16. Generate all combinations

17. Combination sum I, II, III

18. Subsets with duplicates

19. Palindrome partitioning

20. Word search (in grid)

21. Letter combinations of phone number

22. Generate parentheses

23. Restore IP addresses

24. Partition equal subset sum

25. Wildcard pattern matching

26. Regular expression matching

27. Knight probability in chessboard

28. N-Queens II (count solutions)

29. Unique paths III (backtracking)

30. Maximal square with 1s

```

Hard Questions

```cpp

31. Hamiltonian path/cycle

32. Graph coloring

33. Cryptarithmetic puzzles

34. Remove invalid parentheses

35. Word break II

36. Word ladder II

37. Expression add operators

38. Max number of k-sum pairs

39. Split array into fibonacci

40. Strobogrammatic number II

41. Beautiful arrangement

42. Count numbers with unique digits

43. K-th permutation sequence

44. Permutation of string (with duplicates)

45. Combination sum IV (DP + recursion)

46. Target sum (assign + or -)

47. Find minimum time to finish all jobs

48. Matchsticks to square

49. Tiling a rectangle

50. Can I win (game theory)

51. Flip game II

52. Guess the word (interactive)

53. Max length of concatenated string

54. Partition to k equal sum subsets

55. Check if number is sum of powers of three

56. Pyramid transition matrix

57. Maximum score words formed by letters

58. Unique paths in grid with obstacles

59. Path with maximum gold

60. Maximum number of achievements

```

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12. 📈 **DYNAMIC PROGRAMMING**
```

Easy Questions

```cpp
1. Fibonacci numbers (memoization)
2. Climbing stairs
3. Minimum jumps to reach end
4. Maximum sum subarray (Kadane's)
5. House robber I
6. Coin change (minimum coins)
7. Longest increasing subsequence (length)
8. Edit distance (simple version)
9. 0/1 Knapsack basic
10. Rod cutting
```

Medium Questions

```cpp
11. Longest common subsequence
12. Longest palindromic subsequence
13. Shortest common supersequence
14. Minimum insertions to make palindrome
15. Partition equal subset sum
16. Target sum (count ways)
17. Coin change 2 (number of ways)
18. Unique paths in grid
19. Unique paths with obstacles
20. Minimum path sum
21. Triangle (minimum path sum)
22. Maximum product subarray
```

23. Decode ways

24. Word break

25. Combination sum IV

26. Perfect squares

27. Count bits (DP + bit manipulation)

28. Maximum length of pair chain

29. Longest arithmetic subsequence

30. Longest mountain in array

```

### Hard Questions

```cpp

31. Wildcard matching

32. Regular expression matching

33. Edit distance (full)

34. Distinct subsequences

35. Interleaving strings

36. Scramble string

37. Burst balloons

38. Minimum cost to cut stick

39. Maximum profit in job scheduling

40. Best time to buy/sell stock with cooldown

41. Best time to buy/sell stock IV (k transactions)

42. Maximum sum of 3 non-overlapping subarrays

43. Frog jump (stones)

44. Minimum insertions to form palindrome

45. Palindrome partitioning II (min cuts)

46. Word break II (all sentences)

47. Count different palindromic subsequences

48. Minimum distance to type word

49. Maximum score from multipliers

50. Stone game I-VII series

51. Predict the winner

52. Minimum falling path sum

53. Cherry pickup I and II

54. Largest sum of averages

55. Number of music playlists

56. Count all possible routes

57. Minimum difficulty of job schedule

58. Number of ways to stay in place

59. Minimum number of days to eat oranges

60. Count ways to build good strings

61. Maximum value of k coins from piles

62. Maximum number of points with cost

63. Minimum cost to merge stones

64. Remove boxes

65. Strange printer

66. Minimum score triangulation

67. Minimum cost tree from leaf values

68. Super egg drop

69. Minimum XOR value pairs

70. Count of integers with even digit sum

71. Number of ways to divide corridor

72. Maximum path quality in graph

73. Minimum total distance traveled

74. Count increasing quadruplets

75. Maximum employees to be invited

```

📚 **BONUS: SYSTEM DESIGN QUESTIONS (for each DS)**

Arrays/Lists

```cpp

1. Design a music player playlist

2. Implement a photo album viewer

3. Design a task scheduler

4. Implement a leaderboard

5. Design a calendar application

```

Trees

```cpp

1. Design file system hierarchy

2. Implement auto-completion

3. Design organization hierarchy

4. Create a expression evaluator

5. Design a network routing system

```

Graphs

```cpp

1. Design social network connections

2. Implement map navigation (shortest path)

3. Design recommendation system

4. Create a dependency resolver

5. Design network topology

```

Hashing

```cpp

1. Design URL shortener

2. Implement cache (LRU, LFU)

3. Design spell checker

4. Create a distributed hash table

5. Design a search autocomplete

```

Heaps

```cpp

1. Design task scheduler with priority

2. Implement real-time analytics

3. Design emergency room triage

4. Create a stock order book

5. Design a load balancer

```

🎯 **INTERVIEW TIPS**
```

For each problem, be ready to:

1. **Time & Space Complexity** analysis

2. **Edge cases** (empty input, single element, duplicates)

3. **Multiple approaches** (brute force → optimized)

4. **Trade-offs** between solutions

5. **Real-world applications**

Common Patterns to Master:

- Sliding window

- Two pointers

- Fast & slow pointers

- Merge intervals

- Cyclic sort

- In-place reversal

- BFS/DFS

- Topological sort

- Union find

- Trie

- Segment tree

This comprehensive list covers ~500+ problems across all major data structures! 🚀